

When Safe Search Budgets Do Not Transfer Across Graph-ANNS Rebuilds: [Experiments & Analysis]

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Abstract

Rebuilding a graph index can change the search procedure on which an approximate nearest-neighbor search budget was calibrated. We study when a source-build budget retains its recall target on a rebuilt index, how this transfer should be audited, and what target-side recovery costs. ICBA organizes the analysis around build-conditioned risk, isolated selection and certification roles, and explicit search-cost accounting. In a SIFT-100K and Arxiv-Nomic-100K query-profile replay after 5% data refresh and rebuilding, direct TCP pool reuse has empirical target failure rates of 6.69% and 7.24%. Target-selection recalibration reduces these rates to 1.79% and 2.04%, with mean distance-count reductions of 44.39% and 44.79% relative to the registered fixed endpoint. Cross-family and scale analyses extend the evidence beyond this recovery case. The central systems implication is to treat a rebuilt index as a new calibration environment: measure portability, separate observed recovery from a valid certificate, and account for the cost of obtaining target information.

Keywords

approximate nearest-neighbor search, graph indexes, index rebuilding, risk auditing, budget transfer

W3 writing draft. Introduction, Related Work, Theory, and ICBA/TCP are drafted. Later sections remain a structural draft. Numeric exhibits use the W0 frozen macros; Figure 2 specifies the frozen replay, while other outlined exhibits are planned placements. Section 9.1 lists the remaining checks.

1 Introduction

Search quality depends on both the graph index and the budget used to explore it. The index determines which paths a query can traverse, while the budget controls how much of that search is explored. HNSW and Vamana make this tradeoff central to efficient approximate nearest-neighbor search [11, 13]. After rebuilding, a budget selected on the previous graph may execute on a different search structure. Preserving the parameter value does not establish that its recall behavior is preserved. This creates an operational question at the boundary between index maintenance and query serving: *when can an existing budget policy be reused, and when is target-side information worth acquiring?*

The question has two distinct settings. Rebuilding an unchanged vector collection can alter the graph through construction order or randomization. Rebuilding after a data refresh additionally changes the collection and potentially the nearest-neighbor ground truth. Prior work documents construction sensitivity and develops efficient update mechanisms [8, 12]. Adaptive search methods address another part of the problem: allocating work to individual queries using predicted recall or search cost [5, 10, 14]. These capabilities make budget reuse attractive, but its value

still depends on the relationship between the source calibration environment and the target index.

Evaluating that relationship requires more than comparing average recall. A policy can attain a high mean while a subset of queries falls below the requested quality. We study the probability that a query’s recall is below a specified threshold, separately from the expected fraction of missed neighbors. In the primary experiment, failure means $\text{Recall@10} < 0.95$: because recall is discrete, even one missing neighbor is a failure. A 5% risk target therefore asks that at least 95% of queries recover all ten neighbors. This event is stricter than mean recall of 0.95 and must remain explicit when interpreting external methods.

Our organizing insight is that budget portability belongs to a *source policy, target build, and query workload*. ICBA turns this dependence into an audit. It records what information determines each action, separates target selection from certification and evaluation, and distinguishes absolute target failure from additional failure relative to a reference. The theoretical analysis is organized around the observation assumptions under which source information can or cannot support a target decision. The empirical analysis asks how often reuse fails, which evidence supports recovery, and what that recovery costs. TCP provides the recovery case study: it reuses historical query-budget profiles and adjusts their decisions with target-selection information, followed by a certification decision and fallback.

The primary replay covers SIFT-100K and Arxiv-Nomic-100K, with ten target builds per dataset after a 5% mixed delete/insert refresh. Direct TCP pool reuse yields empirical target failure rates of 6.69% and 7.24%. Target-selection recalibration reduces them to 1.79% and 2.04%, while reducing mean distance calculations by 44.39% and 44.79% relative to the registered fixed endpoint. These results concern queries with corresponding old-build profiles; they measure empirical recovery in that replay population. Search savings and a valid certificate for the entire candidate/fallback decision are separate outcomes.

The broader evidence is deliberately layered. External-method bridges examine query-level risk under their supported interfaces, while Vamana and Deep1M analyses examine different graph-family and scale conditions. These blocks retain their own references and failure definitions. Recovery is then evaluated against its acquisition cost: obtaining target labels and profiles can outweigh cheaper searches over a short serving horizon. Thus, an operator needs both a risk assessment and an amortization calculation before replacing direct reuse with recalibration.

We make four contributions:

- (1) A build-conditioned formulation that separates absolute risk, reference-relative transfer loss, and the information available to the budget policy.

Planned exhibit: Source-to-target budget transfer

Show the same query workload passing through source calibration and a rebuilt target graph. Place three decisions at the target: reuse, recalibrate, or fixed endpoint. Mark the location of risk measurement and information-acquisition cost.

Figure 1: Overview placement. Scientific question: which part of a calibrated search decision changes when the graph is rebuilt? This schematic will explain the setting rather than depict an unmeasured effect.

- (2) An ICBA audit workflow connecting query roles, discrete failure events, certification decisions, and search-cost accounting.
- (3) A layered empirical study of budget portability across the registered recovery, method-bridge, graph-family, and scale settings.
- (4) A TCP recovery case study that quantifies the search benefit of target information and its workload-dependent acquisition cost.

Together, these contributions support treating the search policy as part of the index’s versioned operational state, with an explicit decision about reuse after rebuilding.

2 Related Work and Positioning

Graph construction, evaluation, and maintenance. HNSW combines hierarchical navigation with graph search, while DiskANN introduces Vamana as part of a system for large-scale search [11, 13]. Azizi et al. [4] compare graph-search methods and construction paradigms across datasets and scales. Elliott and Clark [8] show that insertion order and intrinsic dimensionality affect HNSW recall. These studies establish that the constructed graph matters to search quality. We examine a different experimental unit: a budget policy transferred from a source build to a target build, with a matched reference used to separate target difficulty from transfer loss. FreshDiskANN [12] supports insertions and deletions without repeatedly reconstructing the entire index. Such maintenance mechanisms motivate distinguishing graph-only rebuilding from refresh-plus-rebuild; the latter is the setting of our primary TCP replay.

Adaptive work allocation and declarative recall. LAET predicts query-specific termination conditions using both static and intermediate-search features [10]. DARTH instead predicts achieved recall during execution and adapts prediction intervals to terminate HNSW or IVF searches at a declarative target [5]. ANNiE estimates the work required to reach a recall target from graph-derived features; its ANNiE-S search uses those estimates directly [14]. Ada-ef models query-to-dataset similarity distributions to choose an HNSW exploration parameter [16]. It also studies incremental updates to its statistics and estimation table, including comparisons with stale and fully recomputed variants after insertions or deletions. Our question concerns the validity and cost of carrying a decision between builds. It applies to adaptive policies as well as fixed budgets, but comparisons must retain each policy’s permitted signals and update procedure. The frozen

DARTH and Ada-ef bridges are evidence about those integrations, rather than a common ranking of all published methods.

Quality guarantees and their target quantities. ConANN applies conformal risk control to IVF search, controlling expected false-negative rate while adapting the number of probed clusters [9]. Its objective is the expected fraction of missed neighbors, whereas our principal event is whether a query falls below a recall threshold. ANNiE’s optional recall-target guarantee directly addresses the fraction of queries reaching the requested recall [14]. Its analysis uses quantile-loss minimization, and its evaluation examines both guarantee deviations and out-of-distribution queries. The DARTH+ preprint also introduces optional probabilistic quality guarantees for HNSW and IVF [6]. Consequently, the distinction of our study is not the existence of a quality guarantee. We focus on which policy and build a claim covers, what information is reused or reacquired, and how the resulting decision behaves after rebuilding. A failure rate computed with our event does not, by itself, refute a guarantee for a different loss.

Calibration, selection, and the role of ICBA. Conformal risk control bounds an expected monotone loss under its calibration assumptions [3]. Learn then Test formulates risk-controlled parameter selection as multiple hypothesis testing [2]. These are statistical foundations for auditing a search decision, rather than guarantees that automatically survive a changed execution environment. ConANN explicitly aligns calibration with index construction and assumes exchangeable query data [9]. In our setting, rebuilding can change the loss induced by the search even when the query population is retained. ICBA makes the target build, failure event, candidate selection, and fallback rule explicit. TCP supplies a concrete query-profile reuse and recovery case within that audit. This separation lets us ask whether target information restores useful search efficiency without conflating empirical recovery, a pipeline-level certificate, and net lifecycle savings.

3 Problem Formulation and Theoretical Lens

Budget transfer has three different statistical objects: a decision made from available observations, query risk on a fixed target build, and coverage over a distribution of builds. We state their guarantees separately. The testing, confidence-bound, and rank arguments below are classical tools applied to the registered search setting, rather than new general statistical inequalities.

3.1 Build-Conditioned Risk and Budget Labels

Let G include the vector snapshot, graph, and fixed search implementation, $q \sim Q$ a query, and $\mathcal{A} = (a_1 < \dots < a_J)$ a finite grid of implementation-native actions. A policy π maps permitted observations to an action. For recall target τ , define

$$\begin{aligned} Z_G(q, a) &= \mathbf{1}\{\text{Recall}@k(G, q, a) < \tau\}, \\ \mathcal{R}_G(\pi) &= \mathbb{E}_{q \sim Q} Z_G(q, \pi(q)). \end{aligned} \tag{1}$$

The arguments describing observations are suppressed in $\pi(q)$, but remain part of the policy definition. Ground truth is computed against the corresponding vector snapshot. In the primary replay, $k = 10$ and $\tau = 0.95$ require all ten neighbors; recovering nine is a failure.

For a source policy π_s and a specified target reference π_t^{ref} , the incremental risk is

$$\Delta_{s \rightarrow t} = \mathcal{R}_{G_t}(\pi_s) - \mathcal{R}_{G_t}(\pi_t^{\text{ref}}). \quad (2)$$

Both absolute risks accompany this difference. A target reference can itself fail, and a refresh-plus-rebuild comparison changes more than the graph alone. Requested action, distance evaluations (NDC), and elapsed time are recorded separately: ordering a_j does not order either realized search work or raw recall.

This distinction also affects budget labels. Define the first passing action $B_G^{\text{first}}(q) = \min\{a_j : Z_G(q, a_j) = 0\}$ and the stable-tail action

$$B_G^{\text{tail}}(q) = \min\{a_j : Z_G(q, a_\ell) = 0 \ \forall \ell \geq j\}. \quad (3)$$

An empty set has value $+\infty$, denoting an unresolved endpoint, not an executable action. A finite stable-tail label establishes success of every grid action above it *on that recorded response*; a first passing label has no such implication when recall is non-monotone. Both are retrospective labels requiring truth. An under-budget event or a censored label is therefore kept distinct from the execution failure in Equation (1).

3.2 What Limited Observations Cannot Resolve

Fix a query and two possible build environments E_0, E_1 . Let O contain everything a decision rule can observe: source profiles, permitted query features, and any target probe transcript. Its laws are P_0^O, P_1^O . An action may encode execution, fallback, or abstention. Let D_i be the acceptable decisions in E_i under a declared risk-cost requirement, and let L_i be a nonnegative loss. Including cost matters: a common expensive endpoint may satisfy safety in both environments while failing the efficiency requirement.

PROPOSITION 3.1 (OBSERVATION-LIMITED DECISION LOSS). *If $D_0 \cap D_1 = \emptyset$ and $L_i(a) \geq \lambda \mathbf{1}\{a \notin D_i\}$ for $\lambda > 0$, every common, possibly randomized, decision rule based on O satisfies*

$$\max_i \mathbb{E}_i L_i(A) \geq \frac{\lambda}{2} (1 - \text{TV}(P_0^O, P_1^O)). \quad (4)$$

PROOF. Use the decision to test the environment: return 0 if $A \in D_0$, and 1 otherwise. Under E_0 a testing error implies $A \notin D_0$; under E_1 it implies $A \in D_0 \subseteq D_1^c$. The sum of expected losses is at least λ times the sum of testing errors, which is at least $1 - \text{TV}(P_0^O, P_1^O)$ by the two-point testing bound [15]. Taking half the sum proves the claim. $\square$

The result identifies an information requirement for avoiding conflicting decisions. It gives no positive lower bound if a common acceptable decision exists or the full observations distinguish the environments. For random queries, the relevant observable includes q jointly with the transcript; a bound on transcript marginals alone is insufficient. Our empirical transfer rates are measured outcomes, not numerical instantiations of Equation (4). In particular, classifier accuracy supplies a lower bound on distinguishability, whereas this loss bound needs an upper bound.

3.3 Fixed-Target Selection and Certification

Fix a target G and a family of M candidate policies. Write $r_j = \mathcal{R}_G(\pi_j)$ and $c_j = \mathbb{E} \text{NDC}_G(q, \pi_j(q))$. Selection estimates $\hat{r}_j, \hat{c}_j$ on its

own query role. A useful action must be both distinguishable from unsafe candidates and separated from the risk boundary.

PROPOSITION 3.2 (MARGIN-BASED SELECTION). *On the event $\max_j |\hat{r}_j - r_j| \leq \varepsilon_R$ and $\max_j |\hat{c}_j - c_j| \leq \varepsilon_C$, retain $S = \{j : \hat{r}_j + \varepsilon_R \leq \delta\}$ and choose $\hat{j} \in \arg \min_{j \in S} \hat{c}_j$. Every retained policy is safe. If j_* minimizes cost among policies with $r_j \leq \delta - 2\varepsilon_R$ and this set is nonempty, then*

$$r_j \leq \delta, \quad c_j \leq c_{j_*} + 2\varepsilon_C. \quad (5)$$

PROOF. For retained j , $r_j \leq \hat{r}_j + \varepsilon_R \leq \delta$. The margin gives $\hat{r}_{j_*} + \varepsilon_R \leq r_{j_*} + 2\varepsilon_R \leq \delta$, so $j_* \in S$. Thus $c_j \leq \hat{c}_j + \varepsilon_C \leq \hat{c}_{j_*} + \varepsilon_C \leq c_{j_*} + 2\varepsilon_C$. $\square$

This is a conditional selection statement relative to the margin-feasible comparator, not a bound against every safe policy. Its estimation event must be justified for the actual estimator. It also does not bound the cost of a fallback invoked after certification rejects the selected policy.

For a policy fixed independently of n i.i.d. target-certification queries, with x binary failures, a one-sided Clopper-Pearson bound [7] is

$$U(x, n; \alpha) = \begin{cases} \text{Beta}^{-1}(1 - \alpha; x + 1, n - x), & x < n, \\ 1, & x = n. \end{cases} \quad (6)$$

Acceptance requires $U \leq \delta$. Here α controls confidence error and δ bounds query failure. With zero failures, $U = 1 - \alpha^{1/n}$, so $\alpha = \delta = 0.05$ needs at least 59 certification queries. This minimum establishes possible acceptance, not high acceptance probability for a near-boundary policy.

PROPOSITION 3.3 (SELECTED-POLICY CERTIFICATE). *Freeze M policies independently of certification outcomes and allocate $\alpha_j > 0$ with $\sum_j \alpha_j \leq \alpha$. Compute $U_j = U(x_j, n; \alpha_j)$ on a common i.i.d. certification sample. Select any policy with $U_j \leq \delta$, or abstain if none qualifies. Then*

$$\Pr\{a \text{ policy is selected and } r_{\text{selected}} > \delta\} \leq \alpha. \quad (7)$$

PROOF. The probability that any $r_j > U_j$ is at most $\sum_j \alpha_j \leq \alpha$. On the complementary event, every accepted policy is safe, regardless of which is selected. Independence among the policy bounds is unnecessary. $\square$

This is the standard simultaneous-testing construction [2]. Target selection may define the candidate family before certification; the statement then applies conditionally on that selection. A fallback belongs in the family or needs a separately justified certificate. The bound controls the probability of an unsafe selection, not the error rate conditional on the procedure selecting something, and it concerns one fixed target rather than an entire build campaign.

3.4 Pooling Across Exchangeable Builds

Historical query profiles provide a different route: pool budget labels across builds rather than certify a policy over target queries. Let m source builds and one target have exchangeable labels $B_1(q), \dots, B_m(q), B_t(q)$ for a fixed query q , with $+\infty$ retained in the order. For a build-marginal error level η , set $r = \lceil (1 - \eta)(m + 1) \rceil$. If $r \leq m$, let $a(q)$ be the r th source order statistic; otherwise abstain. An infinite order statistic also causes abstention.

PROPOSITION 3.4 (BUILD-MARGINAL LABEL COVERAGE). *For $r \leq m$, the preceding rule satisfies*

$$\Pr\{a(q) < \infty, B_t(q) > a(q)\} \leq \frac{m+1-r}{m+1} \leq \eta. \quad (8)$$

If these are stable-tail labels, the same bound holds for finite-action execution failure.

PROOF. Break label ties using independent continuous auxiliary variables. Exchangeability makes the target’s augmented rank uniform among $m+1$ positions. A label strictly larger than the r th source label has augmented rank greater than r , giving the inequality. For stable-tail labels, a finite action at least $B_t(q)$ succeeds by Equation (3). $\square$

The argument is conformal rank coverage with builds as the sampling unit [1]. It holds separately for each fixed query, marginally over the source and target build draw, not simultaneously for all queries or conditionally on a particular target. Nine sources have resolution $1/10$; a non-vacuous 5% bound requires at least 19. Old-snapshot and refreshed-snapshot labels need not be exchangeable. First-passing labels additionally require a deployment-specific monotonicity bridge; averaging non-monotonicity over all grid actions does not supply that bridge for a selected action. These distinctions determine which parts of the theory can be used to interpret the TCP replay.

4 ICBA Workflow and the TCP Recovery Case

ICBA connects a portability measurement to an operational decision. It leaves graph construction and native traversal unchanged. Its inputs are source and target build identities, a query workload, the permitted policy observations, an action grid, a failure event, and a cost model. Its outputs separate direct-reuse behavior, the selected policy, certification status, fallback behavior, and lifecycle cost. TCP is the history-based budget policy used to study recovery within this workflow.

4.1 From a Transfer Measurement to a Decision

An audit first fixes the execution contract: vector snapshot, ground truth, implementation, grid, recall target, and endpoint treatment. The same parameter name across implementations is not a shared unit of work. We therefore retain native actions alongside NDC, recall, and measured timing, and identify unresolved endpoints before summarizing minimum budgets.

Target information then follows three paths. Selection chooses a candidate using its assigned queries. Certification checks the frozen candidate; it cannot retrain or reorder that candidate using evaluation outcomes. Evaluation records the completed decision’s risk and cost. The primary replay has disjoint roles of 500 selection, 500 certification, and 1,000 evaluation queries. Role separation concerns target responses: each query can still require a corresponding source-history profile, whose acquisition must be reported.

The audit preserves a direct-reuse row even when it would fail certification. Otherwise fallback could conceal the transfer failure being measured. For recovery rows, it reports the actual chosen

policy and includes all fallback queries in risk and cost denominators. A certificate rejection is a statement about available evidence, not proof that the policy’s true risk exceeds δ .

4.2 The Frozen TCP Replay Policy

The primary refresh experiment uses the action grid

$$\mathcal{A} = (10, 20, 40, 80, 120, 160, 200) \quad (9)$$

for native HNSW search, with $\tau = 0.95$. For each query and old build, the implementation finds the first grid index meeting the recall target. It stores index J if no action passes. Write this encoded index as $\tilde{j}_g(q)$; its unresolved status is a separate property of the underlying responses. For target seed t , the history-max rule is

$$j_0^{(t)}(q) = \max_{g \in \mathcal{H}_t} \tilde{j}_g(q), \quad \pi_0^{(t)}(q) = a_{j_0^{(t)}(q)}, \quad (10)$$

where $\mathcal{H}_t$ contains the nine old builds other than the target-seed build. Executing the endpoint for an unresolved source query does not make its source label finite-safe; target outcomes are still measured normally.

This rule uses historical responses for the *same query*, including queries assigned to target evaluation. The experiment thus evaluates query-profile reuse after refresh, not prediction of a budget for an unprofiled query. No learned query-to-budget predictor is used in this lane. A fresh workload requires obtaining those profiles or declaring another action for a cache miss.

The historical TCP-HM9-TC variant instead uses stable-tail labels at $\tau = 0.90$, retaining $+\infty$ before routing unresolved queries to the endpoint. We keep that variant distinct from the first-passing, $\tau = 0.95$ refresh replay. Neither a shared name nor nine-source maximum pooling supplies the 5% build-marginal guarantee of Proposition 3.4; the label bridge, pool size, and exchangeability assumptions must each hold.

4.3 Target Recalibration and Fallback

Recalibration changes one global grid shift, not a per-query target label lookup. For $s \in \{0, \dots, J-1\}$, define

$$\pi_s(q) = a_{\min\{j_0^{(t)}(q)+s, J\}}. \quad (11)$$

The selection routine scans shifts in ascending order and chooses the first whose selection-role CP upper bound is at most 0.05. If none passes, it returns shift $J-1$, the endpoint policy. This use of the bound is a selection heuristic, not a simultaneous certificate over scanned shifts. It chooses the first qualifying shift, not the minimum measured-NDC policy of Proposition 3.2; larger shifts need not have monotonically increasing NDC.

The chosen policy π_s is frozen before certification. On the separate certification role, the code computes its individual 95% CP bound and the endpoint’s individual 95% bound. It executes π_s if the former passes; otherwise it executes the endpoint and sets a deployment flag according to the latter. If both fail, the replay still records endpoint outcomes with a failed deployment flag. Such a record is evaluation evidence, not an authorized deployment. The source-audited comparator follows the same decision path with $s = 0$; raw source reuse bypasses fallback for diagnostic purposes.

Frozen refresh replay	
(1)	Encode each query’s first passing old-build action; unresolved responses execute at the endpoint.
(2)	Pool nine source indices by their maximum to form $j_0^{(t)}(q)$.
(3)	On selection queries, scan the global shifts in Equation (11); freeze the first qualifying shift.
(4)	On certification queries, check the candidate and endpoint separately. Choose candidate if accepted, else endpoint; retain a failed flag if neither passes.
(5)	Read evaluation outcomes at the chosen actions. Report risk, NDC, tails, fallback, and acquisition costs.

Figure 2: Executable specification of the TCP recovery replay. The target roles are disjoint; source profiles remain required inputs for each query. Step 4 uses individual bounds rather than a simultaneous 95% certificate.

These two individual bounds do not instantiate Proposition 3.3 at $\alpha = 0.05$: their error allocations sum to 0.10. The primary tables therefore retain empirical risk and fallback behavior without a pipeline-wide 95% claim. A separate historical three-policy implementation allocates $0.05/3$ to TCP, a source-selected fixed action, and the endpoint. That composition matches the simultaneous construction, but uses a different policy/event configuration and does not retroactively certify the refresh results.

4.4 What the Operator Pays For

Given a stored pooled index, applying Equation (11) costs a lookup, integer addition, and grid clipping per query before native search. Pooling m stored labels costs $O(m)$; retaining all labels for Q profiled queries costs $O(mQ)$ storage, or $O(Q)$ for an already pooled policy. These operation counts exclude acquiring the labels. Source profiling requires native searches and exact-neighbor truth, while target selection and certification require their own labeled responses. A first-passing profile can stop at a success; a stable-tail profile needs the remaining registered responses too.

The serving gain is the reduction in total NDC relative to the fixed endpoint over the same queries, not the mean of per-query ratios. The audit records mean, query-pooled p95/p99, and fallback-build counts separately, since a build-level endpoint fallback can change the entire cost distribution. Section 8 accounts for the horizon needed to repay information acquisition. An economical policy must repay those costs under its actual query-reuse rate; a cheap lookup alone establishes no lifecycle advantage.

5 Experimental Methodology

Section purpose: Give each evidence block its own population, unit, and event definition.

Primary recovery block. The current numerical tables use the frozen SIFT-100K and Arxiv-Nomic-100K query-profile replay after a 5% mixed delete/insert refresh and rebuilding, with ten target builds per dataset. This block combines data change with graph change; it does not isolate the effect of rebuilding an unchanged collection. It uses 500 target-selection, 500 certification,

Table 1: Planned setup matrix: fields that must be traceable for each evidence block. This is a reporting schema, not a completed experimental inventory.

Block	Required setup fields
Primary replay	Source/target builds; historical query profiles; grid; role counts; risk event
Method bridges	Published method; adapter changes; permitted observations; matched threshold
Cross-family	Native action semantics; reference policy; censoring; source qualification
Scale	Dataset size; build pairs; query overlap; reference and incremental risks
Economics	Cost unit; shared/sunk inputs; acquisition policy; serving horizon

and 1,000 evaluation queries. The complete setup table will identify source-build history, query IDs and overlap checks, vector snapshot, grid, implementation commit, and hardware for each block. These fields must come from the corresponding manifests before a unified setup is asserted.

Metrics. We report empirical failure rates, mean distance-count gain, sample query-pooled p95 distance counts, and fallback-build counts. Let $\bar{c}_{\text{end}}$ and $\bar{c}_{\pi}$ denote the mean work under the registered endpoint and a policy. Search gain is

$$g(\pi) = 1 - \frac{\bar{c}_{\pi}}{\bar{c}_{\text{end}}}. \quad (12)$$

Negative values mean additional work. A lower sample p95 is a descriptive tail comparison; it is not a statistical noninferiority certificate or a latency measurement.

Uncertainty and dependence. Build-level variation is distinct from query variation. The current gain intervals use the frozen nested-bootstrap implementation, which independently resamples queries within each sampled build. Since query IDs may recur across builds, the dependence structure and intended estimand require reconciliation before assigning a population-level interpretation to these intervals. The final robustness exhibit will report build-wise outcomes, the registered leave-one-build-out analysis, and the uncertainty procedure actually used.

Cross-block comparisons. External-method bridges, Vamana re-analysis, and Deep1M use separate protocols. Their absolute risks are not pooled into a common method ranking. A comparison becomes quantitative across blocks only after matching recall event, source qualification, query population, reference action, and information budget.

6 Budget Portability on Rebuilt Graphs

Section purpose: Establish the phenomenon before discussing recovery.

Table 2 shows the primary contrast. Direct source TCP pool reuse has empirical target failure of 6.69% on SIFT and 7.24% on Arxiv. Both exceed the 5% target, despite reducing mean distance counts relative to the fixed endpoint. Thus, in this replay, low search work and target recall compliance are different outcomes.

Table 2: Frozen primary replay results. Risk is empirical Recall @10 < 0.95 failure; gain is mean distance-count reduction against the registered endpoint. These rows use query-indexed old-build profiles.

Dataset	Policy	Risk (%)	Gain (%)
SIFT-100K	Source pool	6.69	59.51
	Target shift	1.79	44.39
Arxiv-100K	Source pool	7.24	64.21
	Target shift	2.04	44.79

Planned exhibit: Transfer matrix and risk-work trade-off

Panel A: source-to-target risk matrix with reference cells. Panel B: per-build risk versus mean distance counts for source reuse, target recalibration, and endpoint. Include the 5% risk target without treating it as a confidence bound.

Figure 3: Phenomenon placement. Scientific question: is transfer failure systematic or concentrated in a small subset of builds? Per-build frozen inputs are required before plotting.

The target-selection lane reduces the failure rates to 1.79% and 2.04%. Its mean search savings remain 44.39% and 44.79%. This contrast motivates the recovery analysis: target information changes the operating point, but its value must include acquisition costs and the validity of the resulting decision rule.

Mechanism analysis to attach. The source-to-target matrix should show which builds and queries drive failure. Separate index changes, source-policy qualification, target difficulty, and endpoint censoring before attributing a gap to a single cause. A deterministic-contract or margin-based explanation will enter this section only after its event and evidence are aligned with the primary results.

7 Cross-Method, Cross-Family, and Scale Evidence

Section purpose: Extend the phenomenon while keeping protocol differences visible.

Registered method bridges. The DARTH bridge records 22.55% and 23.12% target failure on SIFT and Arxiv, respectively. The AdaEF bridge on Arxiv records 9.52%. These observations belong to the specific registered adapters and threshold settings. The final comparison will state what was retained from the original method and what changed for the transfer test.

Vamana. The aligned post-hoc audit reports 16.86% and 11.37% risk for SIFT and Arxiv, using six source builds, six distinct target builds, 36 source-target pairs, and 750 queries. This evidence adds a graph-family case. Its post-hoc status and source-qualification rule remain visible rather than being collapsed into the primary replay table.

Deep1M. The scale block contains 8 builds, 1000 queries, and 56 directed pairs. Transport risk is 28.86%, while the reference risk is

Table 3: Evidence map across protocols. Values are scoped observations, not a common leaderboard. The event and reference for each row must accompany the expanded experiment description.

Evidence block	Frozen result and interpretation
DARTH bridge	SIFT 22.55%; Arxiv 23.12% target failure in the registered adapter
AdaEF bridge	Arxiv 9.52% target failure in its bridge protocol
Vamana audit	SIFT 16.86%; Arxiv 11.37%; post-hoc cross-family evidence
Deep1M	20.04 pp increment over 8.82% reference risk

Planned exhibit: Layered evidence and boundary cases

Use separate panels for external-method bridges, Vamana, and Deep1M. Display the local event, target/reference pair, and uncertainty unit in each panel. Include negative and nontransfer cases when their frozen provenance is reconciled.

Figure 4: Scope placement. Scientific question: which parts of the phenomenon recur across settings, and which depend on the action class or source/reference definition?

Table 4: Target-selection replay: query-pooled sample p95 distance counts and fallback builds. Each dataset has ten target builds.

Dataset	Target p95	Endpoint p95	Fallback
SIFT	2349.05	2615.00	0/10
Arxiv	2974.10	3252.05	1/10

8.82%. Their frozen difference is 20.04 percentage points, with reported interval [17.29, 22.77]. The reference risk is already above 5%; the difference measures an additional transfer penalty, not failure of an independently established safe reference.

8 Recovery Quality and Lifecycle Economics

Section purpose: Turn the recovery example into an operational trade-off, not a universal algorithm ranking.

Search and tail behavior. For the target-selection lane, the frozen mean-gain intervals are [43.88, 44.91]% on SIFT and [34.72, 50.15]% on Arxiv. Their bootstrap interpretation follows Section 5. Table 4 provides the descriptive tail comparison and fallback counts. The lower observed p95 values refer to distance evaluations; they do not measure GPU/CPU latency or establish statistical tail noninferiority.

A common cost unit. In a distance-count model, let C_{acquire} include the nonshared profile, truth, selection, and certification work, and let $\bar{c}$ denote serving work per execution. For a fixed expected

Planned exhibit: Recovery and amortization

Panel A: build-wise risk, mean NDC, and p95 for the registered actions.
 Panel B: modeled net work versus query executions, with explicit source-profile accounting scenarios and nonamortizing targets retained.

Figure 5: Recovery placement. Scientific question: when does an empirically favorable target policy repay the information needed to obtain it? Model curves must be labeled separately from measured search counts.

workload,

$$C_{\pi}(N) = C_{\text{acquire},\pi} + N\bar{c}_{\pi}, \quad (13)$$

$$N_{\text{BE}} = \frac{C_{\text{acquire},\pi} - C_{\text{acquire},\text{ref}}}{\bar{c}_{\text{ref}} - \bar{c}_{\pi}}, \quad (14)$$

when the denominator is positive and the acquisition difference is positive. Nonpositive acquisition difference means immediate modeled amortization if serving is cheaper; nonpositive serving savings require a separate case rather than a positive break-even claim. Shared costs cancel only when the competing strategies genuinely incur the same cost.

Conditional model outputs. The current dataset summaries report break-even at 138836 query executions for SIFT and 107378 for Arxiv. Zero SIFT targets and 1 Arxiv target have no positive modeled serving savings. These values condition on the source-profile cost treatment and reuse assumptions in the frozen model. Target truth is charged, but old-build truth is not explicitly included as a separate acquisition term. A valid lifecycle interpretation therefore requires its shared or sunk status to be specified symmetrically for the reference strategy.

Operational conclusion to test against the final evidence. TCP recovery is useful when the needed query profiles are available, target recalibration yields an acceptable operating point, and the serving horizon amortizes acquisition work. For a new-query workload, the missing profile-acquisition path changes both deployment and economics. The final text will give these regimes rather than claim that one recovery strategy dominates at every horizon.

9 Reproducibility and Interpretation

Section purpose: State the consequences of the evidence boundaries once, at the level where they change the conclusion.

Evidence lineage. The numerical exhibits are generated from a frozen ledger linked to committed summary tables, per-build records, analysis code, and decision manifests. This writing stage does not regenerate raw query responses. The final artifact description will distinguish numerical consistency checks, replay reproducibility, and independent statistical validation.

Deployment population. The primary TCP result concerns queries with corresponding old-build profiles. Target-role separation does not make those source-side query profiles free or available for an unseen workload. This population definition is necessary both for a deployment claim and for amortization.

Theory and confidence. Section 3 distinguishes observation-limited decision loss, fixed-target certification, and build-marginal label coverage. The empirical tables do not instantiate an upper bound on full-observation total variation. The refresh replay also differs from the stable-tail pooling and simultaneous-certification constructions in its label and decision rules, as specified in Section 4.2.

Implication for index operations. Budget policies should be versioned with the build and workload on which they were assessed. A rebuild can trigger a targeted audit without requiring a universal prescription to retrain every policy. The operator needs three separate answers: the new risk, the cost of a valid decision, and the horizon over which that decision is useful.

9.1 Writing Completion Map

This subsection is an internal writing aid and will be removed from the submission manuscript. Sections 1–4 are drafted, including four propositions with proofs and an implementation-aligned TCP specification. Subsequent stages complete the setup matrix, check the shared-query resampling estimand, and document source-profile acquisition costs. Numerical instantiation of the information bound and a simultaneous 95% certificate for the refresh replay remain unsupported by the frozen evidence. Existing Faiss, deterministic-contract, and larger build-farm results require ledger extension before their headline values enter this draft.

The main paper budget is 12 pages excluding references. Compact proofs are included for W3 review; their final allocation between the main text and a separate short appendix will follow the completed experimental narrative. The remaining exhibit placements reserve space for verified results.

10 Conclusion

Search-budget reuse across graph rebuilds is a measurable transfer decision. The frozen evidence shows target failures under direct source reuse and empirical recovery from target recalibration in the query-profile setting. ICBA connects those observations to explicit risk events, independent query roles, and lifecycle accounting. The resulting lesson is to evaluate portability and recovery together: retained search efficiency, statistical assurance, and the cost of acquiring target information describe different parts of the same operational decision.

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